

Algebra

Home Work 3

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Question 3.1.1

Part 1:

Let $\phi : G \rightarrow H$ be a homomorphism between the two groups G and H . Let $E \leq H$ be a subgroup of H . Prove that: $\phi^{-1}(E) \leq G$ (i.e. $\phi^{-1}(E)$ is a subgroup of G).

Let $K = \phi^{-1}(E) \subset G$. We just need to show that K is a subgroup of G . We apply the usual subgroup tests:

- **Check K not empty:**

From the properties of a homomorphism we know that $1_G \in K$ (and hence K not empty).

- **Check K contains inverses:**

$$k \in K \implies \phi(k) \in E \implies \phi(k)^{-1} \in E \implies \phi(k^{-1}) \in E \implies k^{-1} \in K.$$

These steps follow as E itself is a group and ϕ a homomorphism.

- **Check K stable:**

$$\text{Let } a, b \in K \implies \phi(a), \phi(b) \in E \implies \phi(a)\phi(b) = \phi(ab) \in E \implies ab \in K.$$

□

Part 2:

If $E \trianglelefteq H$ prove that $\phi^{-1}(E) \trianglelefteq G$. Deduce that: $\text{Ker}(\phi) \trianglelefteq G$.

From part (1) we know that $E \leq H \implies \phi^{-1}(E) \leq G$; so we just need to confirm that E normal in H implies that $K = \phi^{-1}(E)$ normal in G . Now take any $g \in G$ and $k \in K$. Then:

$$\begin{aligned} \phi(gkg^{-1}) &= \phi(g)\phi(k)\phi(g^{-1}) = \phi(g)\phi(k)\phi(g)^{-1} && \text{(since } \phi \text{ is a homomorphism)} \\ \implies \phi(gkg^{-1}) &\in E && \text{(since } \phi(k) \in E \text{ and } E \text{ is normal)} \\ \implies gkg^{-1} &\in \phi^{-1}(E) = K \end{aligned}$$

Hence, $K = \phi^{-1}(E)$ must be normal in G .

Question 3.1.3

Given that A is an abelian group and that $B \leq A$. Prove that A/B is an abelian group.

Since A is abelian we know that B is a normal subgroup and that A/B is a group. To show that it is itself abelian let us first look at the definition of the group A/B :

$$A/B = \{aB \mid a \in A\}$$

We now just need to show that the binary operation defined on A/B is commutative.

Let αB and βB (where $\alpha, \beta \in A$) be two elements of this set. Then:

$$\begin{aligned}(\alpha B)(\beta B) &= \{(\alpha b_1)(\beta b_2) \mid b_1, b_2 \in B\} \\&= \{\alpha b_1 \beta b_2 \mid b_1, b_2 \in B\} \\&= \{\beta b_1 \alpha b_2 \mid b_1, b_2 \in B\} && \text{(since } A \text{ is abelian)} \\&= \{(\beta b_1)(\alpha b_2) \mid b_1, b_2 \in B\} \\&= (\beta B)(\alpha B)\end{aligned}$$

□

Question 3.1.5

Let G be a group and G/N a quotient group (and hence N is normal). Show that the order of $gN \in G/N$ is the smallest power of g such that $g \in N$.

Firstly, the order of $gN \in G/N$ is the smallest $n \in \mathbf{N}$ such that: $(gN)^n = N$. Now, this is equivalent to: $g^n N = N$. But this last equation only holds when $g^n \in N$. So, clearly, we have that: $|gN|$ is the smallest positive integer such that $g^n \in N$.

□

Question 3.1.9

Define: $\phi : \mathbf{C}^\times \rightarrow \mathbf{R}^\times$; by $\phi(a + bi) = a^2 + b^2$.

1. Proof ϕ is a homomorphism:

We have two properties that we must prove:

- Check ϕ maps identity to identity.

Now $1_{\mathbf{C}^\times} = 1$ and $1_{\mathbf{R}^\times} = 1$. And we see that: $\phi(1) = 1$ - so this property holds.

- Check $\phi(ab) = \phi(a)\phi(b)$:

Let $a = a_1 + a_2i$ and $b = b_1 + b_2i$. Then:

$$\begin{aligned}\phi(ab) &= \phi((a_1 + a_2i)(b_1 + b_2i)) = \phi(a_1b_1 - a_2b_2 + (a_1b_2 + a_2b_1)i) \\ &= [a_1b_1 - a_2b_2]^2 + [a_1b_2 + a_2b_1]^2 \\ &= [(a_1b_1)^2 + (a_2b_2)^2 - 2(a_1b_1a_2b_2)] + [(a_1b_2)^2 + (a_2b_1)^2 + 2(a_1b_2a_2b_1)] \\ &= (a_1^2 + a_2^2)(b_1^2 + b_2^2) \\ &= \phi(a_1 + a_2i)\phi(b_1 + b_2i) \\ &= \phi(a)\phi(b)\end{aligned}$$

These two properties holding demonstrates that ϕ is indeed a homomorphism.

The kernel of this homomorphism are all those complex numbers in $\mathbf{C}^\times$ whose modulus is 1. This is simply those complex numbers of modulus 1 - i.e. the circle of complex numbers of radius 1 around the origin.

Question 3.1.10

Question 3.1.14

Question 3.1.20

Question 3.1.32

Q_8 is the quaternion group, where:

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\} \quad (3.1.32.1)$$

$$i^2 = j^2 = k^2 = ijk = -1 \quad (3.1.32.2)$$

$$ij = k, \quad jk = i, \quad ki = j \quad (3.1.32.3)$$

- 1) Prove every subgroup of Q_8 is normal.
- 2) For each subgroup find the isomorphism type of its corresponding quotient.